Semantic Voice Compression for LoRa Mesh Networks in Infrastructure-Less Environments

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***Abstract*—Infrastructure-less environments - such as disaster zones, remote rural areas, and mountainous regions - impose severe bandwidth and power constraints that prevent conventional communication systems from supporting human-scale interaction. While long-range, low-power technologies like LoRa offer extended coverage, their limited throughput (11-242 bytes per packet) renders traditional voice transmission impractical. However, through compression and reconstruction via artificial intelligence we have achieved a ˜99.6% compression rate, changing a 946kB file into a 4kB file and effectively reducing our packet count from 7,391 to 32. We demonstrate this through a neural audio codec, EnCodec, which operates natively at the original audio sample rate (48 kHz) and encodes speech into a compact set of discrete token indices that can be transmitted over our peer-to-peer LoRa network, with the model pre-installed at both ends for local reconstruction. Our validation of this AI compression leads us to propose plans to evaluate a more sophisticated model: a generative neural vocoder at the receiving node reconstructs intelligible speech from the compressed semantic tokens, paired with a LoRa mesh network. This system will be tested across rural, urban, and mountainous environments, measuring end-to-end latency, packet loss, semantic similarity, mean opinion score (MOS), and energy efficiency. Compared to raw audio transmission and text-based alternatives, our semantic compression proposal aims to maintain communicative richness essential for emergency response and remote community connectivity. This work establishes a foundation for intelligent, infrastructure-independent communication systems that prioritize intent preservation over raw data fidelity.**

I. Introduction

Environments lacking communication infrastructure, whether due to geographic isolation, disaster, or systemic limitation, often fail to support what we define as human-scale interaction: the effective transfer of intent and knowledge between multiple parties [1]. Modern communication systems are optimized for high-bandwidth environments where raw audio, text, and multimedia can be transmitted with minimal constraints. In infrastructure-less settings, however, strict limits on bandwidth, power, and network topology prevent conventional systems from preserving the nuance, urgency, and semantic clarity required for meaningful communication.

To address this gap, we propose a low-power, intelligent communication system that is designed not only to transmit data efficiently, but also to preserve the essential urgency and communicative richness of human interaction under severe constraints.

In many parts of the world, infrastructure is still limited or nonexistent. Rural villages, disaster-affected regions, and mountainous communities often lack cellular connectivity, making traditional communication methods ineffective. Conventional voice systems such as walkie-talkies also incur substantial energy costs that directly impact battery lifetime [2]. Long-range, low-power technologies such as LoRa help bridge this gap, but most deployments still depend on centralized gateways or rigid network structures that do not adapt well to unpredictable infrastructure-free conditions [3]. These realities motivate the need for autonomous, power-aware communication systems that remain effective in real-world edge environments.

This paper proposes a semantic-aware voice communication system for decentralized LoRa mesh edge networks. Our approach uses semantic compression to encode the intent and knowledge of a message rather than transmitting dense raw audio. Specifically, we employ a dual-stream semantic encoder and a generative neural vocoder to reconstruct intelligible speech while preserving speaker characteristics, prosody, and urgency cues that are often lost in STT-TTS pipelines [4]. By distributing semantic encoders and decoders across mesh-capable edge nodes, the system reduces dependence on centralized infrastructure while remaining compatible with strict LoRa payload and duty-cycle constraints.

Our objective is to design and evaluate this semantic-aware LoRa voice pipeline under realistic deployment conditions. We evaluate across rural, urban, and hilly environments, measuring end-to-end latency, packet loss, semantic similarity, reconstruction quality via mean opinion score (MOS), per-message energy cost, and node-level power usage. Through this shift from raw-data transport to AI-driven semantic exchange, we aim to enable reliable, infrastructure-independent communication where conventional networks fail.

Our key contributions include:

- Integration of semantic compression with LoRa mesh edge networking
- A power-efficient architecture for resource-constrained environments
- A framework for evaluating human-scale interaction quality under strict bandwidth constraints
- A semantic-aware, speaker-preserving compression pipeline utilizing dual-stream semantic encoding specifically optimized for LoRa payload constraints

II. Related Work

A. Human-Scale Communication in Infrastructure-Free Environments

Almeida et al. [5] presented a gateway-free LoRa mesh network on ESP32, demonstrating its design, self-healing mechanisms, and empirical performance.

Mari et al. [6] introduced the MauMe network, a LoRa multi-hop collaborative protocol with a low-cost implementation example. Both works validate the feasibility of decentralized LoRa mesh networks, but do not explore semantic compression or voice communication, which are central to our proposed system.

B. Voice Over LoRa

Text-only systems are prone to high word error rates in noisy disaster zones and can place a heavy cognitive burden on users who must read and interpret instructions during a crisis. Sending text or using STT-TTS removes emotion, prosody, and urgency cues, leading to loss of communicative richness under constrained emergency conditions.

Kagai et al. [7] demonstrated voice transmission through LoRa networks exceeding one kilometer using a novel audio codec, achieving single-digit second latency in urban environments. While their work proves the feasibility of streaming voice data over LoRa, it focuses on relatively dense audio payloads and does not evaluate rural deployments. In contrast, our work uses semantic compression to achieve smaller packet sizes, reducing both latency and transmission time to extend network range. Their results validate that LoRa can support real-time data transmission, reinforcing the viability of our proposed edge mesh network architecture.

Urazayev et al. [4] proposed a LoRa-based voice communication system using speech-to-text (STT) with optional semantic compression via a large language model, achieving a 94% reduction in latency compared to raw audio transmission. While their proof of concept demonstrates significant latency improvements through semantic compression, their evaluation does not examine transmission range or network topology. In contrast, our work investigates semantic compression performance across varying distances within a decentralized edge mesh architecture, addressing the practical deployment challenges of rural, mountainous, and disaster-affected environments.

C. Semantic Compression

SemantiCodec features a dual-encoder architecture: a semantic encoder using a self-supervised pre-trained Audio Masked Autoencoder (AudioMAE), discretized using k-means clustering on extensive audio data, and an acoustic encoder to capture the remaining details. This design not only enables the codec model to capture more semantic information but also encodes the audio signal with fewer token sequences [8].

Traditional DL-based JSCC methods do not focus on the semantic similarities between the source signal and its reconstruction at the receiver and have mostly considered the mean squared error (MSE) or structured similarity metrics (SSIM/MS-SSIM) as the end-to-end measure of distortion [9].

AudioMAE provides the conceptual foundation for our semantic extraction framework [10]. Unlike traditional encoders, AudioMAE's high masking ratio forces the model to learn the underlying global structure and "essence" of the audio rather than local acoustic fluctuations. This allows our edge node system to ignore high-frequency environmental noise and focus on the latent tokens that represent the sender's intent.

D. Potential Drawbacks

1) Latency: Latency remains a major concern in LoRa networks. Raw audio transmission is generally impractical because limited throughput constrains packet payload size, which increases total transfer time for longer messages. Even with compression and mesh forwarding, end-to-end delay remains sensitive to hop count, queueing, and retransmissions. The impact of multi-hop propagation under realistic channel conditions remains an open question that must be validated through field testing.

2) AI Limitations: Deploying semantic compression models on edge nodes introduces multiple technical bottlenecks. First, deep learning inference on resource-constrained microcontrollers can consume substantial energy; reducing radio airtime does not automatically guarantee net battery savings. Second, generative reconstruction can introduce semantic hallucination, where plausible but incorrect fragments are synthesized when input tokens are corrupted by packet loss. To mitigate this risk, reconstruction policies should prioritize intent integrity over acoustic naturalness and fall back to simplified text output when confidence in semantic recovery is below threshold.

3) Duty-Cycle Constraints: LoRa devices are subject to regulatory transmission-time limits. In Europe, duty-cycle operation is governed by ETSI EN 300 220 [11]. In the Americas, operation under FCC rules imposes transmission constraints, commonly interpreted as approximately 400 ms per 20 s window under typical configurations [12]. In a mesh network, relaying another node's traffic consumes repeater airtime budget and can temporarily reduce that node's availability for its own traffic.

4) Collision: As network density grows, overlapping transmissions increase collision probability. Collisions can cascade through the network by triggering retransmissions, raising congestion, and increasing the chance of corrupted or unusable reconstructions. These effects motivate explicit management of channel allocation, medium access behavior, and duty-cycle scheduling.

5) AI Power Draw: Although LoRa radios are optimized for low-power operation, edge AI processing can impose a significant additional energy burden. Sustained deployment therefore requires power-aware model design and runtime optimization, including TinyML-oriented architectures and efficient inference strategies [13], [14].

III. System Design and Architecture

A. System Overview

Our system design targets infrastructure-less communication scenarios where bandwidth, power, and duty-cycle constraints prevent conventional voice pipelines from operating effectively. The architecture combines AI-based compression and LoRa transport in a staged manner: validated point-to-point (P2P) components establish a working baseline, and proposed mesh and semantic modules extend that baseline toward human-scale, infrastructure-independent communication.

B. Validated Components

1) Validated Hardware Setup: Our current validation platform uses two Heltec ESP32S3 WiFi LoRa V3 modules with SX1262 radios for peer-to-peer LoRa communication. We attach SD card readers with microSD cards on both sides to store transmitted files and capture protocol logs for offline analysis and reproducibility. During baseline validation, the AI codec workflow is hosted in Google Colab, which allows rapid iteration of encoding and decoding experiments before migrating the pipeline to fully local edge hardware.

2) AI Compression Baseline with EnCodec: To assess whether AI-based compression can make LoRa voice transmission practical, we validated a neural audio codec as the primary compression module. After evaluating earlier approaches, including feature-based pipelines with self-supervised speech models and mel-spectrogram inversion, we selected EnCodec as the best-performing and most practical baseline [15]. The baseline AI compression and reconstruction flow is shown in Fig. 1.

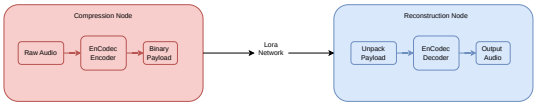

Fig. 1. AI compression and reconstruction pipeline used in baseline validation.

EnCodec is an end-to-end neural codec that maps waveform audio to compact discrete tokens via Residual Vector Quantization (RVQ). The model has three stages: (1) a convolutional encoder that compresses waveform audio to a dense latent stream at 75 frames/s, (2) an RVQ module that maps each frame to one or more integer codes from a 1024-entry codebook, and (3) a mirrored decoder that reconstructs waveform audio from those codes.

For this work, we use the 48 kHz EnCodec model, which operates natively at 48 kHz without resampling. On the transmitting node, audio is fed directly into the encoder. A key implementation detail is segment-level amplitude normalization before encoding: the model removes RMS energy to stabilize encoding scale. The resulting normalization scale is required for reconstruction and is carried in the packet header as a 2-byte float16 value.

At the receiver, decoded token indices and the transmitted scale factor reconstruct quantized latents and synthesize audio directly through the mirrored convolutional decoder, without phase estimation or spectrogram inversion. This makes reconstruction deterministic given tokens and scale, reducing iterative artifact risk observed in earlier baselines.

3) LoRa P2P Transport Baseline: The current validated transport layer is a peer-to-peer LoRa link operating in half-duplex mode, where each node can both transmit and receive. LoRa provides long-range, low-power communication with typical ranges up to 15 km in rural/suburban settings and around 5 km in urban environments [16], [17]. This range comes with strict payload limits, typically 11–242 bytes depending on radio configuration and regulation [18].

To ensure reliable fragmented audio delivery under lossy links, we implement a three-phase session protocol. A session begins with an `AudioStart` packet carrying session metadata (fragment count, codec identifier, sample rate, duration, and CRC16). The receiver returns an ACK to authorize data transfer. The transmitter then sends sequential `AudioData` packets using a fixed 255-byte frame, with 10 bytes reserved for protocol headers and up to 245 bytes for payload. After all fragments, an `AudioEnd` packet carries a CRC32 checksum for end-to-end verification.

ACK status codes distinguish successful receipt, CRC failure, and missing fragments, providing a basis for future retransmission policy. The `codec_id` field supports both raw PCM and compressed audio, enabling controlled comparisons under identical link settings. In initial validation, this pipeline reduced fragment count from 7,391 to 32 for a 946 kB sample, directly reducing airtime, duty-cycle consumption, and transmission energy. Devices operate in the 915 MHz (902–928 MHz) ISM band under FCC requirements [12]. The validated state is single-hop P2P, used as the baseline for per-hop latency, packet loss, and energy benchmarking.

C. Proposed Components

1) Proposed Hardware Integration: Scaling from P2P validation to mesh deployment requires additional LoRa nodes of the same hardware class to support multi-hop forwarding. In the proposed architecture, AI inference is moved from cloud notebooks to local compute (for example, a Raspberry Pi class device or similar edge computer). Hardware integration will establish a reliable interface between LoRa nodes and local AI processing, with candidate pathways including direct

serial links and short-range wireless bridging. Final integration choices will be determined through hardware-in-the-loop testing of throughput, latency, and power consumption.

2) LoRa Mesh Edge Network Extension: Building on the P2P baseline, the next implementation phase extends the transport to a decentralized multi-hop LoRa mesh. In this topology, nodes transmit, receive, and relay packets directly without requiring a central gateway for each transfer [19], [20]. This structure improves coverage, fault tolerance, and locality of processing at the edge [21], and is well matched to low-power semantic audio exchange in infrastructure-free environments.

3) Dual-Stream Semantic Encoder and Generative Reconstruction: Our validated EnCodec baseline currently encodes and decodes full audio tokens without explicit factorization of semantic intent and prosody. The proposed AI module advances this toward a structured dual-stream semantic encoder deployed on mesh edge nodes:

- **Primary Intent Layer:** Discretized semantic tokens carrying core message content
- **Prosodic Layer:** Features preserving urgency, emphasis, and speaker identity

By mapping both streams into compact embeddings, the target transmission budget is reduced from several kilobits to approximately 500–700 bits per message segment, which is more aggressive than current EnCodec operating points (for example, approximately 3,750 bytes for a 10-second clip at two codebook layers). At the receiver, a generative neural vocoder reconstructs intelligible speech from received semantic tokens and local model priors. This proposed generative decompression path is intended to preserve communicative meaning when link conditions approach sensitivity limits.

D. Evaluation Methodology

This evaluation plan has two phases. Phase 1 validates that AI compression can reduce LoRa transmission burden enough to make voice transfer feasible under payload constraints. Phase 2 evaluates full system behavior across realistic deployment conditions.

1) Initial Validation (Completed Baseline): The initial validation was intentionally scoped to pipeline feasibility rather than end-to-end field performance. Testing was conducted indoors at fixed short range (approximately 10 m) using SF7, 125 kHz bandwidth, and the 915 MHz ISM band. Two transmission cases were compared:

- Raw audio: 10 s speech clip at 48 kHz
- Compressed audio: EnCodec token payload packed into `.bin`

Compression reduced file size by approximately 99.6%. Correspondingly, fragmentation dropped from 7,391 fragments (raw) to 32 fragments (compressed), yielding approximately 99.59% faster transfer in this controlled setup. Reconstructed audio remained subjectively intelligible, with speaker identity preserved over the 10 s sample. Observed artifacts were mild ringing and softened consonant onsets, consistent with low-bitrate neural coding at two codebook layers.

TABLE I
TRANSMISSION PERFORMANCE COMPARISON

Metric	Raw	Compressed	Absolute delta (C-R)	Per
Transmission duration (ms)	6221834.00	25344.00	-6196490.00	
Fragment count (fragments)	7391	32	-7359	

The raw-versus-compressed transmission burden comparison used in this baseline is summarized in Table I.

No formal MOS study was performed in this phase; quality was assessed qualitatively for pipeline validation only. EnCodec inference ran in Google Colab (NVIDIA T4 GPU), while LoRa nodes handled fragmentation and radio transport. At present, compression/decompression and LoRa transmission remain separate processing steps; direct on-node AI integration is a planned future step.

2) Proposed Testing (Full System Evaluation): We will evaluate validated and proposed components across three environment classes:

- **Rural:** Open fields
- **Urban:** Obstruction by buildings
- **Hilly/Rugged terrain:** Mountainous regions

Each campaign will collect LoRa transport and AI compression/reconstruction data under comparable channel settings and topology conditions.

3) Evaluation Metrics:

- Bitrate (bps)
- End-to-end latency (ms): $E[L] \approx H \cdot [T_{tx} + T_{queue} + T_{duty} + \frac{P_{loss}}{1-P_{loss}} \cdot (T_{tx} + T_{duty})]$
- Packet loss against hop count
- Semantic similarity score
- Speaker similarity score [22]
- Mean Opinion Score (MOS) [23]
- Energy per transmitted message
- Total energy expended per node

Methodologically, we define a central transmission point and map effective radius per environment. Test traffic includes recordings with varied urgency, duration, and quality. We sweep spreading factor (SF) and bandwidth settings to characterize airtime, reliability, and throughput trade-offs. Geographic coordinates (GPS/GNSS) are recorded for each test position so spatial location can be incorporated into latency and packet-loss analysis.

Our target is approximately 90% packet delivery with greater than 80% MOS. MOS will be evaluated with human listeners through our college using the ITU-T P.800 five-point scale across intelligibility, naturalness, and urgency preservation. We first benchmark point-to-point behavior, then evaluate multi-hop mesh performance across approximately four relay clusters.

4) Why Voice-Semantic Exchange Over Text-Only Alternatives: This system is designed for low-power, human-scale communication where preserving intent and urgency is critical. Text-only alternatives can perform poorly in noisy disaster contexts and can impose cognitive burden on users who must

read and interpret instructions under stress. Likewise, STT-to-TTS pipelines may remove prosody, emotion, and urgency cues. Our approach therefore prioritizes semantic voice reconstruction to retain communicative richness under constrained emergency conditions.

E. System Limitations and Challenges

1) Latency: Latency is a major concern in LoRa networks. Traditional transmission of raw audio over these networks is impractical due to limited throughput, restricting messages to small sizes. While our approach aims to mitigate latency through compression and our mesh network, latency inherently remains a factor, particularly when messages hop between clusters multiple times.

2) AI Limitations: The deployment of intelligent semantic compression at the network edge introduces several technical bottlenecks. The computational overhead of running deep learning models on resource-constrained microcontrollers presents a significant challenge; while we aim to reduce radio airtime, the energy consumed during the inference phase must be carefully balanced to ensure net-positive battery life. Additionally, the reliance on generative reconstruction introduces the risk of "semantic hallucination", where the AI decoder might synthesize plausible but factually incorrect audio fragments. To mitigate this, our system prioritizes intent integrity over acoustic naturalness, implementing confidence thresholds that default to simplified text output if received semantic tokens are too corrupted.

3) Duty-Cycle Constraints: LoRa devices are subject to regulatory duty-cycle limits. In Europe, this is governed by ETSI EN 300 220 [11], while in the Americas, regulations permit transmission for 400ms per every 20 seconds [12]. Within a mesh network, a LoRa device transmitting another device's message takes away from the repeater device's duty-cycle, potentially rendering it unavailable for transmission.

4) Collision: As the network grows, devices will begin to operate on overlapping frequencies, increasing the probability of collisions. Each collision can have cascading effects, causing corrupt messages resulting in retransmission or incorrect reconstructions that completely malform messages. This highlights the importance of managing frequency allocation and duty cycles.

5) AI Power Draw: While LoRa hardware is inherently optimized for ultra-low-power consumption, the integration of high-performance AI models at the edge requires significant computational energy that must be reconciled with the system's limited budget, necessitating a shift toward power-efficient TinyML architectures [13], [14].

IV. Example

V. Conclusion and Future Work

This paper outlines the design and architecture for a semantic-aware voice communication system that operates over a LoRa mesh edge network in infrastructure-less environments. By leveraging a dual-stream semantic encoder and generative neural vocoder, our proposed system aims to compress human voice communication from several kilobits to 500-700 bits while preserving intent, urgency, and speaker characteristics.

Our key contributions include:

1) Integration of semantic compression with LoRa mesh edge networking
2) A power-efficient architecture for resource-constrained environments
3) A framework for evaluating human-scale interaction quality under strict bandwidth constraints

In terms of novelty, while other long-range communication methods exist, most require both infrastructure and power availability. If one or both are unavailable, first-responders and self-reliant rural communities are left without a reliable communication pipeline. Our proposal introduces a semantic-aware, speaker-preserving compression pipeline using dual-stream semantic encoding optimized for LoRa payload constraints, with edge-distributed AI interfaces in a multi-hop LoRa mesh network. The architecture is also duty-cycle-aware for each transmission, reflecting constraints that modern high-bandwidth networks rarely face. Unlike STT-TTS approaches [4], our method preserves prosody and urgency cues that are critical for emergency communication. To our knowledge, this is the first proposal to show how AI-driven semantic exchange can support human-scale interaction in infrastructure-independent scenarios.

Our proposed approach addresses critical gaps in disaster communication and rural connectivity by shifting from raw data transmission to AI-driven semantic exchange. While challenges remain in managing latency, duty-cycle constraints, and semantic hallucination risks, this work establishes a foundation for intelligent, infrastructure-independent communication systems.

Future work will focus on hardware implementation, real-world testing across multiple environments, and comprehensive evaluation of the trade-offs between compression ratio, reconstruction quality, and energy efficiency.

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